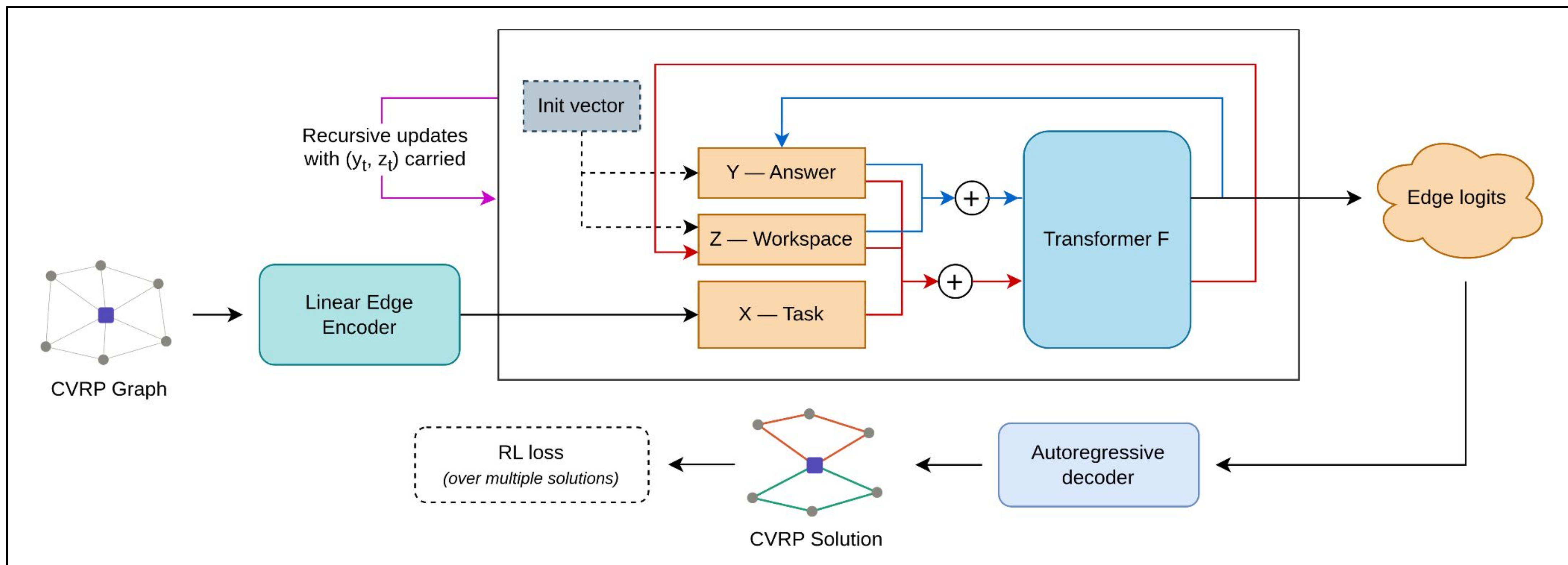


RELAGRAM: Recursive Latent Graph Model for Capacitated Vehicle Routing Problem

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Abstract: We propose RELAGRAM, a novel approach to iterative refinement for graph combinatorial optimization: model improves solution not in the problem space but within a learned latent space. A compact end-to-end Transformer iteratively updates edge-level token representations, which are subsequently decoded to the final solution.



RELAGRAM. A CVRP instance is encoded as edge-token input x ; the answer state y and workspace z start from fixed vectors. The shared Transformer first updates z from $x+y+z$ (red), then updates y from $y+z$ (blue). Both states (y,z) are carried forward (magenta) in a recursive loop. A linear head maps y_t to edge logits, which are decoded into a solution.

Table 1. Gap to the PyVRP reference (%; lower is better). For RELAGRAM, Combined256 uses symmetry, noise, and starts; DeepACO uses 16 ants for 16 iterations.

	RELAGRAM 160k		RELAGRAM 40k	DeepACO 40k
	greedy	256 candidates	256 candidates	256 candidates
CVRP-100	13.40	8.30	9.57	6.08
CVRP-500	22.55	15.64	17.11	17.38
CVRP-1000	29.91	21.65	24.32	23.31

Table 2. Gap to the PyVRP reference (%; lower is better) across recursion depths under greedy decoding for RELAGRAM at 160,000 optimizer updates. The model is trained at depth four.

	depth							best
	1	2	4	6	8	12	16	
CVRP-100	15.29	13.69	13.40	13.35	13.75	14.69	15.49	13.35 @ 6
CVRP-500	25.46	23.74	22.55	22.36	22.44	21.96	22.80	21.81 @ 11
CVRP-1000	33.21	31.58	29.91	28.80	28.75	28.68	29.12	28.45 @ 10

Results: Trained on CVRP-100, the 0.53M-parameter RELAGRAM achieves PyVRP gaps of 13.4% (greedy) and 8.3% (256 candidate search). It generalizes zero-shot to 500 and 1000 instance problems with gaps of 15.6% and 21.7% (256 candidate search).

Depth scaling: Notably, RELAGRAM introduces a new axis for test-time scaling. Trained at depth four, it benefits from 2.5 times deeper recursion on CVRP-1000, reducing the greedy gap by 1.46 percentage points.

Not yet SOTA: However, RELAGRAM does not yet achieve SOTA results. DeepACO, a strong edge-based model, is stronger in most of regimes.

Conclusions: We show that a small latent-recursive model successfully learns to route. It improves beyond its trained depth, presenting promising test-time scaling axis. However, model does not reach SOTA results, and the recursion does not yield decisive gains.

Promising direction is using RELAGRAM for dynamic problems. If RELAGRAM handles an updated instance as well as extra depth, a single model could solve such problems end-to-end, by recurrently processing the last solution and the updated problem.

